Multi-Modal Vehicle Tracking and Counting System Using Optical Flow and Deep Learning

Group 46 — November 23, 2025

1. Group Information and Roles

Arshia Rahim	Student #: 500994106	Detection & Tracking (YOLOv8, Lucas-Kanade)
Ansugan Subramaniam	Student #: 501045133	Data Association & State Estimation (Kalman filter, Hungarian algorithm, track management)
Muhammad Wajeeh Ul Hassan	Student #: 501203009	System Integration & UI (Streamlit interface, visualization)

2. Problem Statement and Approach

Accurate vehicle counting in traffic videos faces challenges from occlusions, similar appearances, varying lighting, and real-time requirements. We combine classical computer vision (Lucas-Kanade optical flow) with deep learning (YOLOv8) for robust tracking, using Kalman filtering for state estimation and the Hungarian algorithm for optimal data association with IoU and appearance features.

3. Objectives

Primary: Achieve >95% counting accuracy on traffic datasets. **Technical:** Real-time processing (>15 FPS), robust occlusion handling (>80% persistence), multi-vehicle support. **Usability:** Intuitive web interface with real-time parameter adjustment. **Research:** Compare optical flow vs. deep learning tracking approaches.

4. Experimental Methodology

Datasets: Custom traffic videos (2+ videos) with varying densities, lighting, angles; manual ground truth annotation. **Metrics:** Counting accuracy (True Count/Ground Truth × 100%), tracking precision (MOTA), FPS, false positive rate. **Setup:** Python 3.8+, OpenCV 4.8+, Ultralytics YOLO, Streamlit; CPU/GPU with CUDA support. **Baseline:** Optical Flow vs. YOLOv8; Parameters: confidence (0.3–0.7), min box size (20–50px).

5. Current Status

Completed: Lucas-Kanade tracker, YOLOv8 integration, Kalman filter, data association, vehicle counter (line/polygon ROI), perspective transform, CLI interface. **In Progress:** Counting accuracy improvements, Streamlit web interface, performance optimization (>15 FPS), enhanced occlusion handling. **Challenges:** Track ID switching in high-density traffic, similar vehicle appearance, adaptive thresholding, memory optimization.

6. Timeline and Milestones

Week	Arshia Rahim	Ansugan Subramaniam	Wajeehul Hassan
1–2	Fix accuracy, improve YOLOv8	Enhance Kalman, track persistence	Adaptive thresholding, UI framework
3	Optimize parameters, multi-scale	Data association improvements	Complete Streamlit interface
4	Test detection on datasets	Benchmark tracking performance	UI testing, integration, visualization
5	Performance optimization (GPU)	Code & algorithm documentation	Multi-threading, final UI polish
6	Final report preparation, presentation slides, project submission (All members)		

Deliverables: Functional web application, evaluation report with metrics, documented source code, final presentation.